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PSYCHEDELIC AND DISSOCIATIVE-ASSISTED THERAPY, CURRENT RESEARCH AND FUTURE POTENTIAL

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Abstract – Psychedelic-assisted therapy (PAT) involving classical psychedelics such as lysergic acid diethylamide (LSD), psilocybin, and ayahuasca, and therapy with dissociatives such as 3,4 methylenedioxymethamphetamine (MDMA) and ketamine, are emerging as effective novel treatment methods for psychopathology. These compounds are of note because while their application as a psychopharmacological agent is novel, many have been studied for decades in other contexts, and so their associated risks are better understood than other newly developed pharmaceuticals. Emerging research is showing that compared to their potential benefits, the short-term risks of taking these substances are negligible, and long-term aversive effects are virtually unheard of. In patients with treatment-resistant affective disorders, PAT is proving to be an effective novel treatment, with a rapid onset and long-term duration of positive effects. Few of these disorders have had designated new treatment methods developed within the preceding decades. The mystical, neurological, and psychological effects of these compounds, in addition to their seemingly high efficacy and long-enduring effects are what make them promising novel options for treating individuals suffering from psychiatric comorbidities. These treatments provide the potential for palliative psychological relief to individuals dealing with psychiatric ailments without having to numb them from their emotional expression. More research needs to be done to understand the potential of these treatment methods, the population groups that could benefit, and how they affect people's receptiveness to other treatments post-treatment with adjunctive psychedelics or dissociatives. Present research warrants additional focus on the elevated ethical concerns present in psychedelic and dissociative research, such as the limitation of informed consent and the exploitation of client patient relationship.

Keywords: psychedelics, psychiatric disorders, treatment-resistant, long-term effects

Psychiatric disorders such as major depressive disorder and post-traumatic stress disorder (PTSD) are incredibly damaging to public health, and if left untreated can be lethal (World Health Organization, 2022). For individuals suffering from severe cases of affective disorders, anxiety disorders, substance use disorders, eating disorders, and traumatic disorders, their condition(s) can produce challenges in multiple complex and interacting aspects of their quality of life, physiological, and psychosocial health (World Health Organization, 2022). But while clinical understanding of these conditions and their effects on personal functioning has become much broader in the past ten years, few novel treatment methods have been developed (Ceskova & Silhan, 2018).

Medical science has been steadily progressing and our understanding of the brain and neurological function is growing due to advancements in imaging technologies. Yet mental health practitioners are still relying on updated versions of treatment methods that are decades old (Ceskova & Silhan, 2018). These treatment methods often require weeks to months of repeated administration to achieve desired effects, and their efficacy is questionable (Gomez et al., 2018). In the case of typical pharmacological agents, common chronic adverse physiological and psychological side-effects are well established (Edinoff et al., 2021). For those who do not respond to these first line treatments, there are few other options currently available that do not involve considerable somatic intervention such as deep brain stimulation, and electroconvulsive therapy. As a result, there exists an urgent need to develop new treatment methods that address existing weaknesses. PAT and dissociatives seem to be of heightened interest in this

context, partly because there are so few novel treatment methods available for treatment-resistant patients (Howes et al., 2022). Furthermore, for those struggling with interacting comorbid psychiatric conditions one single course of treatment often does not provide adequate benefits, and so more intense programs involving multiple treatment approaches are required (Kelly et al., 2012). While mid-twentieth century trials of psychedelics produced evidence for the treatment of psychiatric conditions (Aday et al., 2019), public opinion of psychedelic usage became increasingly unfavorable, including their use in a scientific context. Consequently, research into treatment methods involving psychedelics or pharmacological agents with acute psychoactive effects was stunted and has only begun to re-emerge in a serious capacity within the past two decades (Tupper et al., 2015). The research that is emerging surrounding psychotherapy with adjunctive treatment with drugs such as classical psychedelics and dissociatives is of particular interest as it addresses some crucial weak points in most modern frontline treatment methods: *a)* the long latency period between the beginning of treatment and emergence of positive therapeutic results, *b)* the lack of effective treatment options for those unresponsive to modern forms of treatment, and *c)* the lack of effective treatment options for those suffering from comorbid and interacting psychiatric conditions.

Classical Psychedelics

Classification, Subjective Profile, and Potential Mechanisms

The group classical psychedelics include drugs such as LSD, psilocybin-containing mushrooms, and N,N-diethyltryptamine (DMT). They are some of the most frequently studied psychedelics in a therapeutic context. Their mechanisms of action, efficacy in therapeutic treatment, similar low levels of adverse effects, and comparable changes in perception and subjective experience make them able to be grouped together and loosely generalized into the group of classical psychedelics (Leger & Unterwald, 2021). As a group of drugs which produce an altered state of consciousness, they best represent the therapeutic effects and pharmacological characteristics of psychedelics. All act as potent agonists of 5-HT serotonin receptors, particularly 5-HT2A receptors, which have been implicated in the regulation of depressive behaviors (Nautiyal & Hen, 2017). Since the serotonin system plays a large role in mood regulation (Berger et al., 2009), this could help to explain the acute changes in mood and sense of well-being people often experience during a psychedelic trip, as well as longer-term changes in emotion and affect expression. Compared to dissociatives, classical psychedelics more readily produce vivid perceptual hallucinations such as visual and auditory changes. In addition, they make users more sensitive to sensory information such as light, sound, and touch and provoke feelings and perceptions of abstract mysticism, spiritual, and transformative subjective experiences. Patients often report that their experience taking classical psychedelics had a significant positive impact on them, listing it as one of the most profound and memorable experiences of their lives (Agin-Liebes et al., 2020; Andersen et al., 2020) when allowed to elaborate on treatment experience and effects during follow-up.

Classical psychedelics appear to increase neuroplasticity, synaptogenesis, and spinogenesis. The 5-HT2AR serotonin receptor, with which psychedelics readily bind, has been implicated in the mediation of cortical neuroplasticity (Vargas et al., 2023). A broader potential mechanism is neuroplasticity through the stimulation of increased levels of brain-derived neurotrophic factor (BDNF). As an endogenous compound it is important for brain function and synapse formation, particularly in memory-focused areas such as the hippocampus (McPhee et al., 2020) where the creation of new synapses or long-term potentiation of existing synapses is vital to maintaining proper function. Classical psychedelics appear to have a particularly high binding affinity to receptors which trigger the release of neurotrophic factors, far exceeding that of other antidepressants (Moliner et al., 2023). A second mechanism could be alterations in cortical glutamate levels, which, as a secondary function, cause the release of neurotrophic factors, such as BDNF (Muttoni et al., 2019). Changes in neural networks and inter-neural communication could explain some of the long-term therapeutic effects of psychedelic-assisted therapy. For instance, increased top-down regulation by the prefrontal cortex of the default mode network, a series of regions and pathways implicated in automaticity, and the limbic system, which plays a major role in emotion, seems to reduce habitual negative rumination and autonomic threat response in patients with affective disorders (Carhart-Harris et al., 2017). Increased top-down regulation allows greater mediation by the prefrontal cortex of specific behavior and thought-related brain regions, possibly explaining the myriad of constructive changes detected in patients treated with classical psychedelics in a therapeutic setting. The increased density of dendritic spines produced by classical psychedelics (Ly et al., 2018; Shao et al., 2021) could be responsible for the increased usage of endogenous neurotransmitters. As a result, classical psychedelics

could achieve a similar outcome as SSRIs of increasing monoamine usage, albeit through a different mechanism of action. An additional result of classical psychedelic administration is a lasting increase in global brain integration and resting-state functional connectivity (Daws et al., 2022), along with diminished activity in stress response-related areas such as the amygdala (Carhart-Harris et al., 2017; Muttoni et al., 2019) possibly through the increased top-down regulation by the prefrontal cortex.

Current Research, Common Methods and Features

Common features of classical psychedelic research are *a)* various validated psychopathology scale measurements administered in a pretest-posttest design, *b)* some number of psychotherapy sessions before, between, and after instances of psychedelic intervention to build rapport, establish pathological baselines, and integrate features of the psychedelic experience, and *c)* a comfortable space with eyeshades and calming music provided where the psychedelic experience takes place in the presence of professional clinicians (Andersen et al., 2020; Davis et al., 2020; Goodwin et al., 2022; Griffiths et al., 2016; Leger & Unterwald, 2021; Shnayder et al., 2023). These last two features are vital to patient well-being. With all psychoactive drugs, set (personality, present mindset, and expectations) and setting (environmental conditions, such as physical surroundings and the presence of other people) are important in the drug-taking experience, and are essential with drugs such as psychedelics, which can readily produce a wide variety of vivid hallucinations and acute changes in emotional and psychological state.

To minimize adverse effects and maximize therapeutic results, the set and setting were consistently designed to be comfortable and familiar, with features fairly consistent across studies to provide a stable environment for the psychedelic administration session to take place. In order to facilitate positive therapeutic outcomes, patients are frequently encouraged to engage in relaxed introspection and personal exploration of the psychedelic experience. Therapists are in the room or available nearby to guide the patient through any discomfort that might occur during treatment but are usually instructed to not interfere with or try to manipulate the patient's experience. The goal is to allow the psychedelic experience to take place naturally without strict direction from clinicians. When examining the patient's subjective experiences during psychedelic intervention, multiple studies note the positive correlation between the patient's peak perception of a deep and transcendent mystical experience and the effectiveness of treatment quantified as the effect size between instances of measurement (Andersen et al., 2020; Griffiths et al., 2016), forming a correlation which could explain some of the dose-dependent efficacy common in the current literature. Consequently, there could exist an overlap in the mechanisms that produce an altered state of consciousness and those that facilitate positive therapeutic effects. When psychedelic treatment is over, integrative psychotherapy sessions take place with the goals of discussing the subjective perceptions of the psychedelic experience, as well as any emotional and pathological changes being experienced post-psychedelic intervention. Studies vary in duration and number of therapy and psychedelic sessions, but most conclude with some talk therapy sessions and follow-ups to discuss lasting changes and establish the progression and longevity of treatment results.

Efficacy and Treatment Outcomes

Determining exact efficacy from modern studies of classical psychedelic's therapeutic profile can be challenging due to factors such as small subject groups, variability in treatment design and therapy type, lack of or unclear blinding procedures, narrow subject recruitment criteria, and lack of a control condition. However, many clinical and experimental studies (Davis et al., 2020; Goodwin et al., 2022; Griffiths et al., 2016; Shnayder et al., 2023), in addition to modern meta and systematic-analyses (Andersen et al., 2020; Haikazian et al., 2023; Ko et al., 2023; Leger & Unterwald, 2021; Muttoni et al., 2019; Schimmers et al., 2022), find strong therapeutic results from baseline measurements in multiple areas of psychiatric dysfunction. In a large-scale multi-center randomized double-blind trial by Goodwin et al. (2022), 233 participants with treatment-resistant depression were recruited into a study assessing the antidepressant effects of COMP360, a synthetic psilocybin formulation. Results demonstrated that the highest dosage group of 25mg (n=79) in conjunction with psychological support significantly improved clinician and self-assessed ratings of depression at week three, and reductions from baseline depression levels measured via self-assessment remained by week twelve.

Clinicians working in these studies frequently report marked improvements in patients' psychological well-being, emotional profile, and behavior, including in studies where clinicians rating participants were unaware of the substance administered (Griffiths et al., 2016). Studies frequently rely on clinician and/or self-administered scale assessments. Patients, particularly those facing a diagnosis of terminal or life-threatening illness, often report and demonstrate significant positive changes in more general aspects of psychological health.

Subjects show improvements in constructs such as optimism, spirituality, positive self-image, sense of connection with others and their community, life purpose, sense of peace, positive life-outlook, mortality acceptance, and overall quality of life (Agin-Liebes et al., 2020; Griffiths et al., 2016). One participant noted, "It has given me a different perspective on my life and has helped me to move on with my life and not focus on the possibility of cancer recurring. I try not to hold onto or stress unimportant things" (Agin-Liebes et al., 2020, 162)

Psychedelic-assisted therapies utilizing classical psychedelics produced rapid-acting and sustained positive therapeutic effects in those suffering from affective disorders and substance use disorders (Fuentes et al., 2020) and nicotine addiction. Scores on depressive and anxiety measurement inventories, rates of substance abstinence, personal function, and overall quality of life consistently demonstrate some significant level of dose-dependent efficacy. One meta-analysis of seven studies by Perez et al. (2023) found a significant correlation between the dosage given and level of therapeutic effects for the treatment of depression, where fifty percent of effects were observed at 10.13 mg/70 kg, and ninety five percent of effects were observed at 41.14mg/70 kg. Most significant is the consistent findings of the sustained duration of positive therapeutic effects resulting from a single or multiple psychedelic interventions in studies that have long-term repeated measurement protocols. Although there are still questions about the ideal number of drug interventions, a meta-analysis done by Leger and Unterwald (2021) suggests that repeated psychedelic treatment sessions produce significantly greater efficacy than single-dose treatment.

Efficacy and duration of effects appear strong even in patients characterized as being treatment-resistant (Goodwin et al., 2022), who are a commonly selected cohort in studies examining PAT. Many patients continue to demonstrate large effect sizes from baseline measurements up to and over six months post-treatment. A meta-analysis by Andersen and colleagues (2020) found that psychedelic-assisted therapy was associated with clinically significant reductions in heavy alcohol usage at all assessment points over a 32-week follow-up period. Reductions in depressive symptom scores of around 40-60% were observed six months post-treatment. A follow-up by Agin-Liebes et al. (2020) of cancer patients who received psilocybin-assisted therapy found sustained significant therapeutic effects at the 3.2 and 4.5-year marks. These patients continued to have reduced scores in areas such as anxiety, depression, hopelessness, and demoralization and overwhelmingly reported personally meaningful and positive changes. Other studies have found positive results with anxiety disorders and disordered eating after a single or multiple treatment sessions with classical psychedelics (Leger & Unterwald, 2021; Peck et al., 2023).

The sustained therapeutic effects of classical psychedelics and PAT after a single or small number of treatment sessions more broadly contribute to the interpretation that there exist underlying biological mechanisms and neural changes occurring as a result of PAT methods. These mechanisms likely extend beyond acute changes such as monoamine levels in the brain, the typical mechanism of modern pharmacological intervention. A trial by Carhart-Harris et al. (2021) suggests that two treatments with psilocybin-assisted therapy could be more effective than a continuous daily administration of the modern SSRI escitalopram. However, their trial failed to achieve statistical significance in part due to limited sample size. Longevity of effects may be mediated by an interaction between the neurological effects of psychedelics, combined with integrative therapy sessions. Neurological alterations might have enough of an effect on mood and motivation to enhance therapeutic engagement to a level that would be expected from non-treatment-resistant patients. In addition, the subjective alterations produced by psychedelics might allow for a deeper and richer exploration of patient experiences to facilitate optimal treatment outcomes. If the positive therapeutic effects of psychedelic therapy persist for the duration that the current research suggests, it is likely that neurological alterations contribute to creating and sustaining long-term improvements in mental health.

Adverse Effects

Adverse effects from classical psychedelics appear to be of low concern in a professional therapeutic context as psychedelics appear to be well tolerated by patients (Andersen et al., 2020; Griffiths et al., 2016; Haikazian et al., 2023; Ko et al., 2023; Muttoni et al., 2019; Perez et al., 2023; Schimmers et al., 2022; Shnayder et al., 2023). Psychedelics are not seriously reinforcing or addictive (Johnson et al., 2018) in a controlled therapeutic setting. Commonly reported side effects are fatigue, nausea, headaches, perceived changes in body temperature, elevated heart rate and blood pressure, and a heightened sense of psychological discomfort, abnormality, or anxiety (Bender & Hellerstein, 2022). Recognizing that anxiety during treatment could be unavoidable in achieving positive therapeutic effects is important. Some level of discomfort might be inevitable for patients during the process of facing and processing their psychological issues, and so might not qualify as an adverse effect by definition.

Moreover, these symptoms can be dealt with and managed by therapists and staff present during experimental treatment and in follow-up integrative sessions. Side effects are typically transient, subsiding by the end of or shortly after psychedelic treatment.

Considering the potential benefits of adjunctive therapeutic treatment with psychedelics, the present evidence suggests an acceptable risk-benefit profile. More serious adverse effects such as self-harm, suicidality, and sustained changes in psychological well-being, cognition, and physical health are extremely rare, according to a meta-analysis of adverse effects by Bender and Hellerstein (2022). In the few cases where severe adverse effects were found, they were determined to not be a result of psychedelic treatment. However, this must be taken into context because those who are considered high risk, such as those with active psychotic conditions or are considered at risk of self-harm and suicidal ideation are generally prevented from inclusion in study participation through strict screening and recruitment criteria.

Dissociatives

Classification, Subjective Profile, and Potential Mechanisms

Dissociatives differ from their classical psychedelic counterparts by not as readily producing vivid hallucinations and abstract mystical experiences. While some hallucinations still might occur, they are not nearly as consistent and abstract as with classical psychedelics. Dissociatives do still enhance perceptual sensitivity to light, sound, and touch and similarly affect neurotransmitter systems such as serotonin, dopamine, and norepinephrine (Mitchell et al., 2021; Sessa et al., 2019; Tedesco et al., 2021). But unlike classical psychedelics, dissociatives are characterized by producing subjective experiences where the user feels as if they are disconnected from their body and are not present.

A prominent dissociative, MDMA, is known for enhancing feelings of social connection, compassion, mindfulness, and empathy, as a result promoting deep emotional explorations between oneself and others (Tedesco et al., 2021). One theorized neurological mechanism is MDMA's positive effect on oxytocin signaling (Kirkpatrick et al., 2014), a neuropeptide/peptide hormone associated with supporting ingroup bonding. An increase in neural oxytocin concentration could help make the purposeful recollection and communication of traumatic memories during psychotherapy sessions more tolerable. As a result, these prosocial characteristics are what make MDMA such a prime candidate for treating anxiety and traumatic disorders, particularly in treatment-resistant individuals, because MDMA has the potential to inhibit impulsive fear response and diminish the "characteristic feelings of social detachment and hypervigilance in PTSD [patients]" (Hoskins et al., 2021, 2). Consequently, MDMA promotes a cognitive and socioemotional state conducive to patient-therapist communication, facilitating engagement with traumatic memories and encouraging post-traumatic growth and achievement of fear extinction (Gorman et al., 2020; Hoskins et al., 2021; Mitchell et al., 2021; Tedesco et al., 2021).

Ketamine is the drug that least resembles the psychoactive and therapeutic profile of the other drugs described. Neurological mechanisms of therapeutic effects are believed to be a modulation of glutamate system signaling through the disinhibition of cortical GABAergic interneurons by antagonizing their presynaptic N-methyl-D-aspartate receptors (Abdallah et al., 2016; Browne & Lucki, 2013; Floriano et al., 2023; Lazarevic et al., 2021; Matveychuk et al., 2020). Disinhibition is the inhibition of neurons which inhibit other neurons, thus enabling the increased activation of those neurons. An additional mechanism is an increase in dendritic spine growth (Lazarevic et al., 2021; Wu et al., 2021) due to an increase in BDNF as a result of increased glutamate signaling in the prefrontal cortex (Abdallah et al., 2016).

Current Research, Common Methods and Features

MDMA and ketamine therapies are further along in the treatment development process than are classical psychedelics. Ketamine is approved for usage by practitioners both for its traditional usage as an anesthetic and its new antidepressant usage at licensed clinics. In 2017, MDMA was given breakthrough therapy designation for the treatment of PTSD by the Food and Drug Administration to accelerate treatment development. As such, studies with MDMA are homing in on key treatment areas and study design. Ketamine is most prominently used for its antidepressant effects in those who are resistant to treatment. Typically given either intranasally or intravenously for fast drug absorption, the therapeutic effects are rapid. Less common in ketamine studies and treatment are regular integrative therapy sessions. Ketamine treatment seems to lean more on the purely neurological effects to facilitate treatment outcomes, rather than using it to amplify the effects of other treatment methods.

Studies on MDMA are more similar to those done with classical psychedelics, with structured pretest-posttest protocols, integrative therapy, and drug intervention making up the course of treatment, with frequent sample sizes of about sixty participants (N=18-92) (Gorman et al., 2020; Mitchell et al., 2021; Nicholas

et al., 2022; Ot'alora G et al., 2018; van der Kolk et al., 2024; Wolfson et al., 2020). Many modern studies of MDMA-assisted therapy make use of "A Manual for MDMA-Assisted Therapy in the Treatment of PTSD" developed by the Multidisciplinary Association for Psychedelic Studies. The manual outlines various aspects of therapist selection, creation of a safe physical setting, supportive protocols, preparation, aspects of study and therapy conduction, integrative follow-up design, and other design features to facilitate study methodology that can be more easily compared (Multidisciplinary Association for Psychedelic Studies et al., 2017). Patients in experimental groups often undergo multiple sessions of MDMA treatment, sometimes in two closely occurring dosages. The second dosage being supplemental, normally contingent on patient well-being and subjective state (Tedesco et al., 2021).

The purpose of many MDMA trials with treatment-resistant PTSD patients is to elevate subjects' direct engagement with trauma-focused talk therapy. Since these forms of therapy rely on getting patients to recall traumatic memories to help process them and achieve fear extinction, patients often have trouble actively engaging with these memories without experiencing traumatic anxiety, which may lead to a breakdown in active communication and patient engagement. What MDMA-assisted therapy seems to do is achieve a "window of tolerance" (Mitchell et al., 2021, 1031), where cognitive and emotional engagement with traumatic memories becomes more manageable. The window of tolerance might be most open during the active effects of MDMA, but positive changes in emotional processing, fear management, and openness to social interaction that seem to make talk therapy more tolerable appear to persist into the integration sessions weeks after MDMA intervention (van der Kolk et al., 2024). MDMA seems to contribute to the creation of a prime candidate for trauma-focused talk therapy. Patients given MDMA are, in ideal circumstances, engaged emotionally, highly social, and open to recalling and discussing their experiences, but without undergoing a fear response that pushes them away from therapy or causes defensive disengagement during treatment.

Efficacy and Treatment Outcomes

Ketamine appears to have rapid-acting therapeutic efficacy in treating affective disorders, including in treatment-resistant patients (Boudieu et al., 2023; Drozdz et al., 2022; Floriano et al., 2023; Nikolin et al., 2023), with efficacy varying marginally depending on the number of interventions, dosage, and route of administration. Depressive symptoms are decreased substantially for a period of a few days, with effects peaking around the 24-hour mark. However, the antidepressant and anxiolytic effects of ketamine do not appear to persist nearly as long as MDMA and classical psychedelics. Symptoms typically return to near baseline levels within a few weeks post-treatment (Boudieu et al., 2023; Drozdz et al., 2022; Matveychuk et al., 2020; Schimmers et al., 2022), necessitating follow-up administration to achieve therapeutic benefits. With ketamine treatment, fewer studies have required repeated integrative psychotherapy in addition to drug administration, which could account for the lack of long-term therapeutic outcomes (Schimmers et al., 2022). A meta-analysis by Drozdz et al. (2022) examined studies where ketamine was combined with some form of psychotherapy to treat any type of physical or psychological ailment. Their findings suggest that adjunctive therapy could increase the duration of ketamine's anxiolytic and antidepressant effects, but currently it is unclear whether therapeutic benefits could be extended to the duration seen in research with MDMA or classical psychedelics. The disparity in sustained long-term effects could be because of the differences in induced neuroplasticity and synaptogenesis between ketamine and classical psychedelics. However, a discrepancy could still be beneficial because not all patients can fully submit to the rigorous and involved course of treatment necessary in therapy with psychedelics or alternative dissociatives. Consequently, some patients could benefit from treatment with a drug such as ketamine that appears to have a less abstract subjective experience and provides strong short-term relief after a single intervention. Additionally, ketamine does not readily produce the swath of socioemotional changes that make MDMA a prime candidate in a dual psychoactive therapy treatment model, where one modality of treatment interacts with and enhances the other.

MDMA-assisted therapy appears to be efficacious for treating PTSD and conditions typically comorbid with traumatic disorders. Many studies have affirmed the substantial effect sizes seen in experimental groups using various versions of the Clinician-Administered PTSD Scale (CAPS), a widely used tool for diagnosing and assessing PTSD (Weathers et al., 2013). Many studies have found MDMA to be extremely beneficial and well tolerated by patients with severe and/or treatment-resistant anxiety or PTSD (Gorman et al., 2020; Hoskins et al., 2021; Mitchell et al., 2021; Ot'alora G et al., 2018; Tedesco et al., 2021). MDMA's emotional and sociability-enhancing properties appear to magnify the benefits of therapy and make active engagement more manageable for patients who must repeatedly recall, discuss, and

recontextualize traumatic events. Moreover, these factors also appear to improve rates of patient dropout (Tedesco et al., 2021), which are notoriously high in traumatic therapies (Mitchell et al., 2021) primarily because they are emotionally demanding. A placebo-controlled phase 3 trial of ninety patients found that MDMA-assisted therapy significantly improved measures of emotional intelligence, self-compassion, and feelings of self-capacity in patients with PTSD (van der Kolk et al., 2024). Such factors are crucial in allowing patients to overcome the negative effects of trauma on emotional openness and self-determination and work with clinicians to help overcome traumatic experiences. MDMA seemingly makes the healing and rebuilding process of therapy for victims of trauma more tolerable, justifying its designation as a breakthrough therapy.

Similar to classical psychedelics, therapy with adjunctive MDMA administration produces long-lasting effects, with most studies finding benefits months to a year after treatment ceases. A long-term follow-up of sixteen patients with chronic treatment-resistant PTSD found that "mean CAPS and IES-R [impact of events scale-revised] scores were not statistically different from their 2-month (short-term) mean scores" (Mithoefer et al., 2013, 31). This long-term follow-up was conducted between 17 to 74 months after the final MDMA session. Most subjects reported specific lasting benefits from study participation, such as general well-being (89%), fewer nightmares/flashbacks/intrusive memories (68%), increased ability to feel emotions (68%), and reduced anxiety (63%).

Adverse Effects

Overall, dissociatives seem to be well tolerated by patients in a controlled therapeutic setting (Azari et al., 2023; Boudieu et al., 2023; Ot'alora G et al., 2018). Most adverse effects are brief and nonsevere. Common experiences are dry mouth, teeth clenching, jaw tightness, dizziness, headaches, fatigue, perceived changes in body temperature, nausea, changes in appetite, and temporary increase in anxiety (Mitchell et al., 2021; Tedesco et al., 2021). Exceedingly rare are reports of serious adverse effects such as increased suicidal ideation.

As with classical psychedelic treatment, it should be expected that elevated levels of emotional vulnerability and mood intensity during dissociative treatment could lead to discomfort for patients. However, as long as treatment and clinician protocols are designed to account for these potential changes, their risks can be mitigated, and emotional vulnerability can be utilized to assist in therapeutic mechanisms where appropriate. In considering the potential benefits of adjunctive dissociative treatment with therapy, the observed adverse risks do not present abnormal levels of concern to patient wellbeing.

Current Research Limitations and Risks

However, there are limitations on what the current body of research can tell us. The limited sample size may negatively affect the generalizability of the current data. While ketamine is being used more widely, and MDMA research is progressing, our understanding of classical psychedelics in a therapeutic setting is still limited by small sample sizes. Studies do not always include true control groups or complete cessation of other regularly taken pharmaceutical agents. Integrative talk therapy methods are not entirely consistent or well documented across studies, although as the field moves towards more phase 3 trials, improvements in both sample group size and methodological rigor will be seen. Current sample groups and participating clinicians are usually disproportionately white, so more research involving groups from various ethnic, racial, and socioeconomic backgrounds is necessary to bolster research conclusions. There also exists great concern about potential bias in the current research (Hovmand et al., 2023) arising from the possibility of accidental unblinding of participants and clinicians during intervention due to the strong acute perceptual effects of many psychedelics and dissociatives. Blinding participants to their placement in an experimental group is challenging when psychedelics and dissociatives produce such strong subject effects. Nevertheless, this problem could be accounted for with more rigorous blinding procedures, active placebos, improved participant screening, and studies including multiple dosage levels. Another source of bias could be the improper introduction of spiritual or religious language on the part of the clinician during post-intervention therapeutic followup (Johnson, 2020). Just as in non-psychedelic forms of psychotherapy, discussion of these topics can be appropriate, but must be initiated by the participant as to not break with proper therapeutic scope and invalidate potential qualitative insights. Once again more rigorous therapeutic protocols are the clear solution. Established in most studies with varying dosage levels are the dose-dependent therapeutic effects of psychedelic and dissociative therapy, which decreases concern over functional unblinding affecting validity.

Conversely as it pertains to risk, most patients who are considered at high risk of serious adverse complications such as those with a history of psychosis, self-harm, and suicide attempts are screened out of studies with psychedelics or dissociatives. The likelihood of discovering a substantial risk as it pertains to

physiological or psychological complications resulting from a single psychedelic or dissociative intervention is low. These substances have been tested for many decades both in controlled studies looking to investigate potential neurotoxicity and in observational work examining regular recreational drug users. However, there are still some questions as to their risks in patients experiencing disruptive psychiatric issues and their usage in a therapeutic setting in combination with various forms of psychotherapy. While a controlled professional setting makes adverse effects more manageable, having patients continue to focus on uncomfortable or disruptive subjective experiences poses a potential risk, especially if post-treatment follow-ups do not adequately address patients' needs. In addition to the potential physiological risks that must be considered, ethical and therapeutic risks could be of elevated concern to participants during psychedelic and dissociative treatment. This is as a result of the intensity of the treatment, the vulnerable state it can leave participants in during drug intervention creating the opportunity for sexual exploitation by clinicians, the level of expertise needed to perform the therapeutic component of the protocol, and the limitations of informed consent during experimental intervention elevated by the emotional effects of psychedelics and dissociatives (Anderson et al., 2020; Johnson, 2020; Sznitman et al., 2024). Improved informed consent procedures and clinician training procedures, clearer guidelines and protective measures can all address these issues, which require special consideration during the formulation and review of research protocols.

Possibilities for Future Research

Questions remain within the modern psychedelic and dissociative research, one being in establishing prime candidate groups who would benefit from these novel treatment methods. Most clear are candidates who have been diagnosed with comorbid psychiatric problems. The development of a problem such as substance use disorder rarely develops on its own, instead often following some other diagnosis and/or life-altering event. New research should examine how to broadly improve psychiatric function to maximize the therapeutic outcomes in multiple areas of dysfunction. Psychedelic and dissociative treatments will not replace the role of modern pharmaceuticals, such as SSRIs. However, studies could examine the longer-term effects of psychedelic and dissociative therapy on treatment-resistant patients. Even if adjunctive treatment with therapy does not eliminate pathological symptoms in all patients on its own, does it make patients who were previously unresponsive to various treatments more receptive to those same treatments? Consequently, psychedelic and dissociative therapy would be more akin to surgery, an intense and involved process that both reduces dysfunction directly and primes the patient for a more effective long-term process of healing. Given classical psychedelics' and dissociatives' myriad of neurological alterations, there could exist potential for changes in treatment receptivity for treatment-resistant patients. Studies should focus on controlling blinding methods to diminish expectancy effects. However, does the presence of positive language around psychedelic or dissociative treatment in unblinded clinical trials improve treatment outcomes compared to neutral language? It will become increasingly relevant to control exact language used during the research process as the general public becomes more familiar with modern psychedelic and dissociative research separate from the currently stigmatized view of psychoactive drugs. Do certain types of talk therapy interact differently with particular drugs and patient psychiatric dysfunction? If so, can talk therapy be tailored to create treatment plans and protocols particularly well suited to the needs of each patient? At what point in the number of treatment sessions are diminishing returns seen?

Future studies should make room for deeper qualitative data, particularly the direct reports and testimonies of experimental participants. Their insights into the way psychedelic and dissociative therapy affects the course of their treatment journey are of great value in establishing a clear profile of these substances, refining existing treatment methods, and guiding the objectives of future research. Finally, any forms of future research should seek to strengthen blinding procedures, ethical therapeutic protocols, clinician guidelines, participant screening procedures, informed consent procedures and participant protections. If the area of psychedelic and dissociative science is to move forward and become a truly legitimate field, it can only do so safely and properly if it abides by all of the guidelines and protocols which have made modern scientific research possible. While it can be challenging due to the inherently sensational and intriguing nature of these novel substances, greater focus must be put on ensuring scientific and ethical rigor in exploration of these substances' therapeutic potential. "Despite the association between psychedelic use and ego dissolution, grandiosity can loom large with initial psychedelic experiences, leading even conservative individuals to become wildly enthusiastic about the potentials of psychedelics to heal and transform" (Anderson et al., 2020).

Conclusion

Psychedelic and dissociative therapy are novel approaches to treating psychological ailments. They integrate and build upon existing treatment methods while having the potential to achieve outcomes beyond the capabilities of existing treatments. In addition, they possess potential to provide elevated benefits in terms of onset and duration of therapeutic effects and the breadth of conditions with which they can provide relief. Many studies have found strong efficacy for patients with chronic treatment-resistant psychiatric disorders. Psychedelics and dissociatives seem to be well tolerated in a therapeutic setting with minimal adverse consequences for patient health and function. Many appear to have stimulating effects on neuroplasticity and neural wiring conducive to the treatment of psychiatric disorders. More research is needed to bridge the gaps in present research and should be directed toward the goal of developing practical treatment plans for individuals with comorbid dysfunction and resistance to existing treatment methods. Greater emphasis should be put on ensuring participant safety, proper clinician practice, stronger blinding procedures and methodological rigor, and tempering excited expectations about the future of the field.

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